Milestone 3 Documentation

Project Title: AI-Powered-Enhanced EHR Imaging & Documentation System
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Date of Submission: [26/09/2025]

Objective

The objective of this module is to **automate clinical documentation and ICD-10 coding** using Generative AI. In real-world healthcare, doctors spend significant time on routine documentation and coding. This module aims to reduce that burden by:

1. Automatically generating professional clinical notes from structured patient data.

2. Mapping diagnoses to ICD-10 codes through automation, either via lookup tables or AI-driven coding.

This module ensures that structured data can be transformed into usable **clinical documentation** and **standardized medical codes**, which is crucial for both patient care and healthcare system operations.

Relation to Milestones

Milestone 1: Data Collection & Preprocessing

- In Milestone 1, the team focused on collecting and cleaning **medical imaging data (CT, MRI, X-ray, etc.)** and **EHR data (structured + unstructured, including ICD codes)**.
- The cleaned and standardized datasets form the **foundation** for Module 3. Without this step, the AI would not have reliable data to work with.
- Example connection: The structured EHR dataset prepared in Milestone 1 (like *EHR_cleaned.csv*) provides the patient attributes (age, sex, symptoms, diagnosis) that we now feed into the AI model for generating clinical notes.

Milestone 2: Multimodal Data Alignment

- In Milestone 2, the work shifted towards aligning **different modalities of data**—linking medical images, EHR notes, and ICD codes.
- This alignment ensures that **diagnosis in text** corresponds correctly to **imaging findings** and **coding systems**.
- Example connection: Module 3 extends this by using the aligned patient-level data (diagnosis + ICD codes) and making it actionable through **automated note generation and coding**.

📌 Steps in Module 3

Step 1: Prepare Sample Patient Data

- **What we do**: Since real patient data cannot be used (privacy restrictions), we create **synthetic patient records** with realistic medical cases. These cases include attributes such as age, gender, symptoms, and a diagnosis. The dataset is stored in a CSV/JSON format.
- **Example Case**:
- Age: 45, Male
- Symptoms: Fever, cough, shortness of breath

- Diagnosis: Possible pneumonia
- **Why it matters**: This forms the **input dataset** for AI models in later steps. Having diverse, well-structured cases ensures the AI system generalizes better.
- **Challenge**: Making synthetic data realistic and medically diverse enough to cover common diagnoses and avoid overly simplistic cases.

Step 2: Clinical Note Generation using Azure OpenAI

- **What we do**: Feed the structured patient data into an LLM (like GPT-4 from Azure OpenAI) to generate **professional clinical notes** in proper medical documentation format.
- **Example Output**:
- Patient: 45-year-old male
- Presenting complaints: Fever, cough, shortness of breath
- Assessment: Likely pneumonia based on symptoms
- Plan: Recommend chest X-ray and antibiotics
- **Why it matters**: Automates one of the **most time-consuming tasks for clinicians**—writing notes. This

can improve efficiency and standardization in medical records.

- **Challenge**: Ensuring generated notes are **accurate, medically relevant, and consistent** without hallucinations or irrelevant details.

Step 3: ICD-10 Coding Automation

- **What we do**: Assign ICD-10 codes automatically to the diagnosis field, using either:
  - **Lookup Table**: A pre-prepared CSV/JSON file mapping common diagnoses to ICD-10 codes.
  - **AI-driven Approach**: Asking the AI model to predict the ICD-10 code directly from the diagnosis.
- **Example Output**:
- Diagnosis: Pneumonia
- ICD-10 Code: J18.9
- **Why it matters**: ICD-10 coding is essential for billing, insurance, and hospital record-keeping. Automating this saves clinicians time and ensures **standardized classification**.
- **Challenge**: ICD-10 is vast and complex. Lookup tables can be limited, while AI-driven approaches

may sometimes return approximate or incorrect codes. Careful validation is needed.

Step 4: Integration (Prototype Level)

- **What we do**: Build a prototype pipeline (in Python/Node.js) that:
  1. Reads patient data (CSV/JSON).
  2. Sends it to Azure OpenAI → generates the clinical note.
  3. Matches the diagnosis → ICD-10 code (lookup or AI).
  4. Saves results (patient info + note + ICD-10 code) into an output file.
- **Why it matters**: Demonstrates an **end-to-end workflow**, showing how AI can seamlessly fit into hospital processes for documentation and coding.
- **Challenge**: Integration requires careful handling of APIs, data formats, and error handling so the pipeline runs smoothly.

Step 5: Documentation & Submission

- **What we do**: Prepare the final deliverables:

    - Input file (synthetic patient data in CSV/JSON).
    - Code scripts (commented and reproducible).
    - Output file (patient data + generated notes + ICD-10 codes).
    - README.md (explains the workflow, assumptions, and usage).
- **Why it matters**: Clear documentation ensures the project is reproducible and understandable by mentors, evaluators, and future teammates.
- **Challenge**: Balancing **technical clarity** (for developers) with **clinical readability** (for medical reviewers).

Key Challenges Overall

1. **Synthetic Data Quality** – Creating diverse and realistic cases.
2. **AI Reliability** – Preventing hallucinations and ensuring medical accuracy in notes.
3. **ICD-10 Complexity** – Covering a wide range of codes while maintaining accuracy.
4. **Technical Integration** – Managing input-output flow and API usage without errors.

5. **Documentation Clarity** – Writing in a way that serves both technical and clinical audiences.

✅ Expected Outcome

By the end of Module 3, the team will have a **working prototype system** that:

- Takes structured patient input.
- Generates standardized clinical notes automatically.
- Assigns ICD-10 codes automatically.
- Produces a consolidated, well-structured output file.
- It ultimately demonstrates how Generative AI can reduce **documentation workload**, improve **coding efficiency**, and support **real-world healthcare workflows**.